Rembert Daems

Fundamental research in computer vision and stochastic dynamics.
Broad experience in applied research in computer vision and robotics.
Co-founder of a medical device start-up, with an AI diagnosis system.
I love sharing knowledge and connecting the right people and ideas.

EDUCATION AND EXPERIENCE

Ph.D. Researcher at Ghent University, Belgium

MARCH 2021 - PRESENT

Research in neural stochastic differential equations, diffusion models, computer vision, deep learning, 3D vision, robotics, differentiable physics, differentiable rendering, agriculture applications. See selected publications below.

Co-founder at Hippo Dx

FEBRUARY 2020 - PRESENT

Hippo Dx developed an automated skin prick test device (SPAT), enabling consistent and precise allergen testing and freeing precious medical professionals time.

My main responsibilities are the camera and custom lighting system, image processing and automated diagnosis with deep learning. We recently raised €4.4M for the commercialization of the SPAT.

AI Lead at Octinion

MARCH 2018 - FEBRUARY 2021

Computer vision and deep learning technical expert, introduced deep learning in the company. Coaching and guidance for colleagues working on deep learning projects. Developed custom deep learning models for a wide array of computer vision applications, deployment on edge devices and in the cloud. Implemented robots with visual simultaneous localization and mapping (SLAM) for autonomous greenhouse navigation.

Innovation Engineer at CNH Industrial

SEPTEMBER 2016 - FEBRUARY 2018

Computer vision and deep learning expert in the innovation team. Developed computer vision algorithms for various agricultural applications . Developed a camera system for acquiring data in the field autonomously. Coached several internship students in computer vision and deep learning projects.

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Born June 11th, 1991.
Married, father of Karel (6), Rozanne (4) and Korneel (2).

github.io/rdaems
linkedin.com/in/rembert-daems
github.com/rdaems
twitter.com/rembertdaems

SKILLS

Research, computer vision, deep learning

JAX, PyTorch, TensorFlow

Python, C/C++

Linux, ROS, PostgreSQL

HTML, CSS, JavaScript, Node.js

LANGUAGES

Dutch (fluent)
English (proficient)
German, French (basic)

MISCELLANEOUS

Board member AIG (Alumni Engineers Ghent)

Family, (jazz) piano, cycling, reading, renovating our home and always too many side projects.

Bachelor and Master of Science in Electromechanical Engineering

2011 - 2016

Ghent University, Belgium

Master Dissertation:
Fluoroscopic Evaluation of Knee Kinematics

Knee motion analysis of patients with a knee replacement to ultimately improve the design of knee prostheses. In cooperation with the multidisciplinary team at the Ghent University Hospital Biomechanical Lab.

Service:

Student representative

President of BUK (regional student association)

SELECTED PUBLICATIONS

Daems, R., Opper, M., Crevecoeur, G., Birdal, T. (2024). Variational Inference for SDEs Driven by Fractional Noise. **Spotlight** (top 1.2%) at the International Conference on Learning Representations (**ICLR**) 2024.

Daems, R., Taets, J., wyffels, F. & Crevecoeur, G. (2022). KeyCLD: Learning Constrained Lagrangian Dynamics in Keypoint Coordinates from Images. Oral presentation (top 1.6%) at the Machine Learning and the Physical Sciences Workshop at **NeurIPS** 2023.

De Roovere, P., **Daems, R.**, Croenen, J., Bourgana, T., de Hoog, J. & wyffels, F. (2022). CenDerNet: Center and Curvature Representations for Render-and-Compare 6D Pose Estimation. Proceedings of the European Conference on Computer Vision (**ECCV**) Workshops 2022.

Daems, R., & wyffels, F. (2020). Unsupervised Orientation Learning Using Autoencoders. Presented at the Differential Geometry meets Deep Learning Workshop at **NeurIPS** 2020.

Daems, R., Victor, J., De Baets, P., Van Onsem, S., & Verstraete, M. (2016). Validation of three-dimensional total knee replacement kinematics measurement using single-plane fluoroscopy. International Journal of Sustainable Construction and Design, 7(1).